QP CODE: 25900286

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Elective (DSE) - MG3DSECHE200 - BASIC ANALYTICAL CHEMISTRY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Instruction to the Examiners:

- *This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.*
- *Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.*
- *Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.*

Part A
Short Answer Type Questions
Answer any 5 questions (Each question carries 4 marks)

1. **State the observation when silver nitrate is added to a solution containing chloride ion in the presence of nitric acid.**

White precipitate of AgCl formed, Reaction: $Ag^+ + Cl^- \rightarrow AgCl(s)$.

(Correct explanation with reaction give 4 marks)

2. **What is the difference between eliminating oxalate and fluoride ions in terms of the reagents used?**

Oxalate removed by heating so that oxalate is decomposed to carbonate and CO_2.

Fluoride removed using Conc. HCl. By this HF is removed **(Correct answer: 4 marks)**

3. **Discuss the importance of evaluation of hazards before starting an experiment.**

Ensure safety in the laboratory, prevents accidents and chemical exposure, risk assessment, hazard awareness, to adopt control measures, emergency response , helps select proper PPE and ventilation. **(Any four valid points – 4 marks)**

4. **Describe the procedure of titrating oxalic acid solution with standard $KMnO_4$**

Pipette out oxalic acid into conical flask., Add dilute H_2SO_4 and warm to 60–70°C.

Titrate with $KMnO_4$ until permanent pink colour persists.

Reaction: $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$. **(Correct procedure -give 4 marks)**

5. **What is gravimetric analysis?**

Quantitative analysis, analyte determined from mass of the precipitate and its chemical composition , involves precipitation, filtration, drying and weighing.

Eg:-Used for determining Fe as Fe_2O_3, Ba as $BaSO_4$ **(Any three valid points: give 4 marks)**

6. **How can Rf value be used to identify compounds?**

Rf=Distance travelled by the component from the origin/ Distance travelled by the solvent (solvent front)

Each compound has a unique Rf under same conditions., Rf of unknown compared with known sample **(Correct answer-give 4 marks)**

7. **Discuss the main instrumental components involved in gas chromatography**

Carrier Gas System – Inert gas (e.g. He, Nitrogen) transports the sample through the column.
Injector – Introduces the sample into the carrier gas stream.
Column & Oven – Separates components based on volatility and interaction with stationary phase.
Detector – Detects and records separated components as peaks (e.g. FID, ECD,TCD) **(Basic components -give 4 marks)**

Part B

Short Essay Type Questions- Answer any 5 questions (Each question carries 7 marks)

8. **Discuss the rules for rounding off numbers, providing examples.**

Drop all digits in a number not significant,
Number to the right of significant numbers-
if between 6-9-round up one unit, if between 0-4 leave unchanged, if 5 round to nearest even number
In addition and subtraction -same number with minimum number of decimal places
In multiplication and division-as same number with least number of significant digits
(Proper explanation with any four valid points-5 marks ; Example-2 marks)

9. **State the major precautions to be taken while weighing chemicals on an electronic balance.**

Use clean, dry container, avoid placing chemicals directly on pan, close balance doors before reading, limit sample's exposure to air, samples can be dried and weighed so that results will not be affected by humidity, Do not overload the balance **(Any five valid points give 7 marks)**

10. **Discuss the problems that can be arised if glassware is not properly cleaned before experiments?**

Causes contamination of chemicals or reagents, Contamination alters concentration, Interference in titration results, Formation of unwanted precipitates, Reduced accuracy and reproducibility. **(Any 5 valid points give 7 marks)**

11. **What are the advantages of conductometric titrations compared to other titration methods?**

No indicator required, suitable for colored/turbid solutions, accurate end point detection , applicable to a variety of reactions like weak acid/base titrations, redox, precipitation titration etc**.(Any 4 valid points with explanation give 7 marks)**

12. **Why is it important to use a filter paper of appropriate pore size during filtration?**

Large pore size-Fine particles may pass through, leading to a turbid filtrate, Small pore size-Filtration becomes slow or may clog. Right pore size ensures optimal flow rate without compromising the removal of unwanted solids, it balances speed and precision, especially important in time-sensitive or high-throughput workflows, inappropriate pore size can lead to loss of analyte or contamination of the filtrate (**Any five valid points give 7 marks)**

13. **List the steps involved in a simple distillation process.**

Heating a liquid mixture to separate components based on differences in boiling points, Pour the liquid mixture to be distilled into the round-bottom flask. Add boiling chips to promote smooth boiling and prevent bumping, Gradually heat the flask using a heating mantle or water bath, Monitor the temperature with the thermometer placed at the distillation head, As the mixture reaches the boiling point of the more volatile component, it vaporizes, Vapors enter the condenser, which is cooled by circulating water. The vapor condenses back into liquid form. **(Explanation with steps-give 7 marks)**

14. **Describe the principle of reverse phase chromatography and how it differs from normal phase chromatography.**

Liquid chromatography technique used to separate compounds based on hydrophobicity

Reverse-phase Chromatography-Stationary Phase: Non-polar (typically silica modified with long-chain alkyl groups like C18), Mobile Phase: Polar solvents, Mechanism: Polar compounds elute faster because they interact less with the non-polar stationary phase. **Normal-phase chromatography** -Stationary Phase: Polar (e.g., unmodified silica or alumina).Mobile Phase: Non-polar solvents (e.g., hexane, chloroform).Mechanism: Polar compounds are retained longer due to strong interactions with the polar stationary phase. Non-polar compounds elute faster **(Principle -2 marks, Any 2 differences: 5 marks)**

Part C

Essay Type Questions (Answer any 1 question -Each question carries 15 marks)

15. **Discuss the concepts of mean, variance, and standard deviation,including their** mathematical formulas and applications in statistical analysis and data interpretation

Mean ($\bar{x}$): Average of all observations, Formula: $\bar{x} = \Sigma x / n$.

Represents central tendency of data.

Variance (s^2): Measure of data spread around mean, Formula: $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$.

Standard Deviation (s): Square root of variance., Formula: $s = \sqrt{s^2}$.

Indicates precision and reliability of measurements.

Smaller s → higher precision; larger s → greater variation.

Used in data interpretation, error analysis, and quality control.

(Mean, variance, and standard deviation definition with mathematical formulas: 3+3+3; Any one application for each: 2+2+2 marks)

16. **Briefly discuss the basic principle and procedure of (a) Direct titration(b) Indirect titration (c) Back titration (d) Replacement titration**

a) Direct: Titrant reacts directly with analyte. Metal ion in solution reacts directly with EDTA to form a stable complex.

Take the metal ion solution (e.g., Ca^{2+}, Mg^{2+}),Add suitable buffer (pH ≈ 10) and indicator (Eriochrome Black T).Titrate with standard EDTA until color changes from wine red to blue.

Calculate metal ion concentration from volume of EDTA used

b) Indirect: Used for estimation of ions such as Sulphate, Phosphate etc. which do not directly react with EDTA. These ions first converted to metallic salts (like $BaSO_4$) by adding excess metal ion solutions (Ba^{2+}) and then excess metal ion is titrated with EDTA.

c)Back titration: Used when direct titration is slow or incomplete. Excess EDTA is added to the metal solution; Unreacted EDTA is titrated with a standard metal solution.

d) Replacement titration: Type of complexometric titration in which the metal ion of interest cannot be directly titrated with EDTA. It is replaced by another metal ion that can react with EDTA, and the amount of the displaced metal ion is then determined by titration. Each type involves distinct procedural steps.

(Principle and procedure of (a) Direct titration b) Indirect titration (c) Back titration (d) Replacement titration- give 15 marks; Principle and procedure of any two with correct explanation give 10 marks; any one give 5 marks)